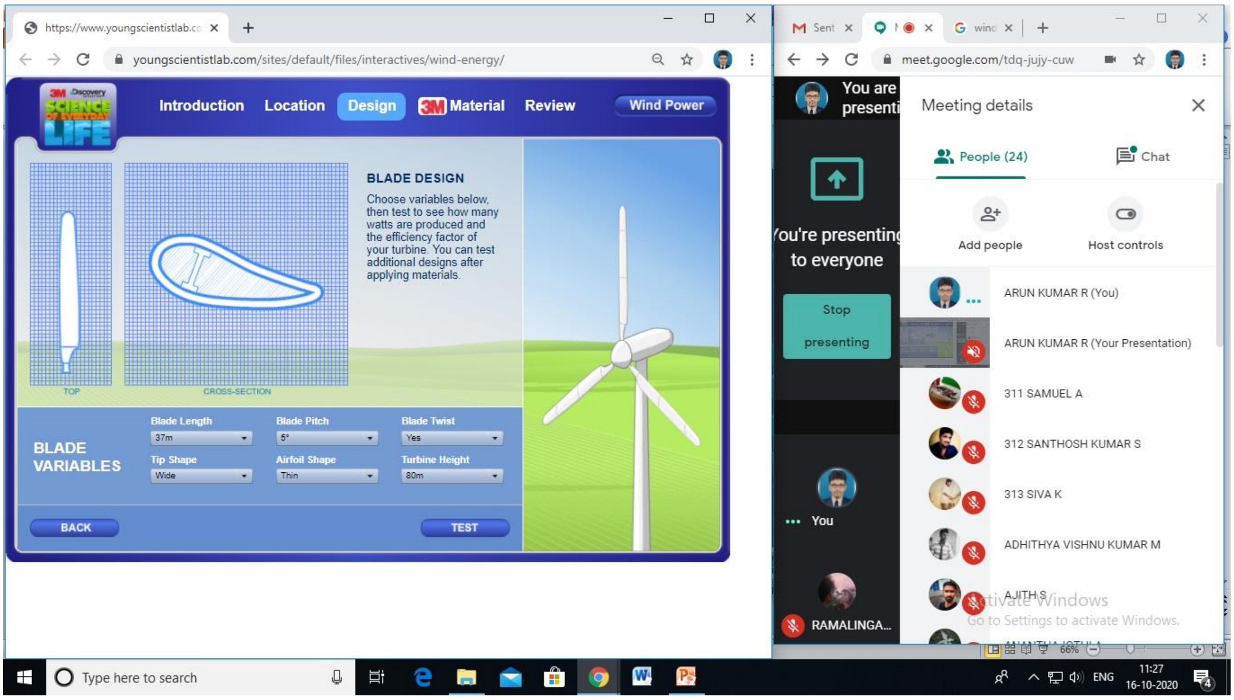
RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

RIT- IITM PALS VLAB Initiative

Virtual Laboratory Session Handling Details

Name of the Faculty	:	Mr.R.Arun Kumar
Academic Year	:	**2020 – 21 (Odd Semester)**
Subject Code and Title	:	ME8072 Renewable Sources of Energy
Year and Branch	:	VII Semester B.E. Mechanical
Topics Covered	:	Site selection criteria in wind farm installation
Name of the Laboratory	:	Young Scientist Laboratory
Website Link	:	https://www.youngscientistlab.com/sites/default/file interactives/wind-energy/
No.of Students participated	:	24
Assignment given if any (If Yes share the uploaded folder link)	:	Yes



RAMCO INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering
Academic Year: 2020 – 2021 (Odd Semester)

ASSIGNMENT SHEET

Degree, Semester & Branch: VII Semester B.E. Mechanical Engineering 'A' & 'B'
Course Code & Title: ME8072 Renewable Sources of Energy
Name of the Faculty member: Mr.R.Arun Kumar, AP/Mechanical

Assignment No.: II **Date:** 09.10.2020

Questions:

1. Submit a report reflecting the learning outcomes attained through ***'Student Solar Ambassador Workshop 2020'*** attended on October 2nd, 2020. The report should include following details:
   a. One page learning outcome
   b. Proof of participation (either screenshot or certificate) **[CO2, L3]**

(Or)

2. Write an inference of what you have understood by using the ***'3M Wind Energy Virtual Laboratory'*** developed by Young Scientist Laboratory. Vary anyone parameter by fixing other seven parameters as constant. Generate three outputs and write what you have inferred. Include the following in the report:
   a. Inference of learning
   b. Screenshot of the results **[CO3, L3]**

<table>
<tr><th>Criteria</th><th colspan="4">Ratings</th><th>Marks</th></tr>
<tr><td>Timely Submission</td><td>Excellent
Submitting on or before time
(4 marks)</td><td>Good
Submitting a day delay from deadline
(3 marks)</td><td>Satisfactory
Submitting within 3 days from deadline
(2 marks)</td><td>Need Improvement
Submitting after 3 days from deadline
(1 mark)</td><td>4</td></tr>
<tr><td>Acknowledgement</td><td colspan="2">Excellent
Reference sources are properly cited
(3 marks)</td><td colspan="2">Unsatisfactory
No proper citation
(0 marks)</td><td>3</td></tr>
<tr><td>Content of the assignment</td><td>Excellent
Impressive, unique content with proper screenshot of attendance or output
(7 to 10 marks)</td><td colspan="2">Satisfactory
Content with grammar error and improper inference from the outcome
(3 to 6 marks)</td><td>Need Improvement
Poor write-up without proper inference
(0 to 2 marks)</td><td>10</td></tr>
<tr><td>Presentation</td><td>Excellent
Impressive presentation with properly aligned content along with title page
(3 marks)</td><td colspan="2">Good
Pleasing presentation with improperly aligned content
(2 marks)</td><td>Need Improvement
Poor presentation without title page
(1 mark)</td><td>3</td></tr>
<tr><td colspan="5" align="right">Total Marks</td><td>20</td></tr>
</table>

Note: Students those who haven't attended the workshop can prefer the assignment on Virtual Laboratory and those who have attended will get an option of either preferring assignment 1 or 2.

CO2: Student will be able to discuss the method of power generation from Solar Energy

CO3: Student will be able to explain the method of power generation from Wind Energy

L3: Apply

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	-	-	-	-	3	-	3	3	2	3	-	3
CO3	-	-	2	1	3	-	3	3	2	3	-	3

Signature of the Faculty member **HOD**

RENEWABLE SOURCES OF ENERGY

ME8072

ASSIGNMENT – 2

3M Wind Energy Virtual Laboratory

Aswin V R

953617114012

VI Mech A

A)

From this assignment, I have learnt how the performance and efficiency of a wind turbine will be affected due to any change in the dimensions of a wind turbine blade or the shape of the tip or the turbine height.

According to this design plan, wind turbine farm can be implemented on three different types of locations (Plains,hills and offshore). While comparing the complexity of maintenance, farms on plains were less complex compared to hills, and complexity of farms on hills were less compared to farms offshore.

But despite the complexity of maintenance, wind turbine farms present on plains powered less number of homes compared to hills and offshore. Wind turbine farms present off-shore powers the most number of homes.

The dimensions I have given in the following screenshot gave the best efficiency factor and powered the most number of homes.

From my design, despite the complexity in maintenance, Offshore farms powers the most number of houses at top efficiency factor in low watt/unit cost.

B)

10/16/2020 https://www.youngscientistlab.com/sites/default/files/interactives/wind-energy/

3M Wind Energy Design Results

Design 1

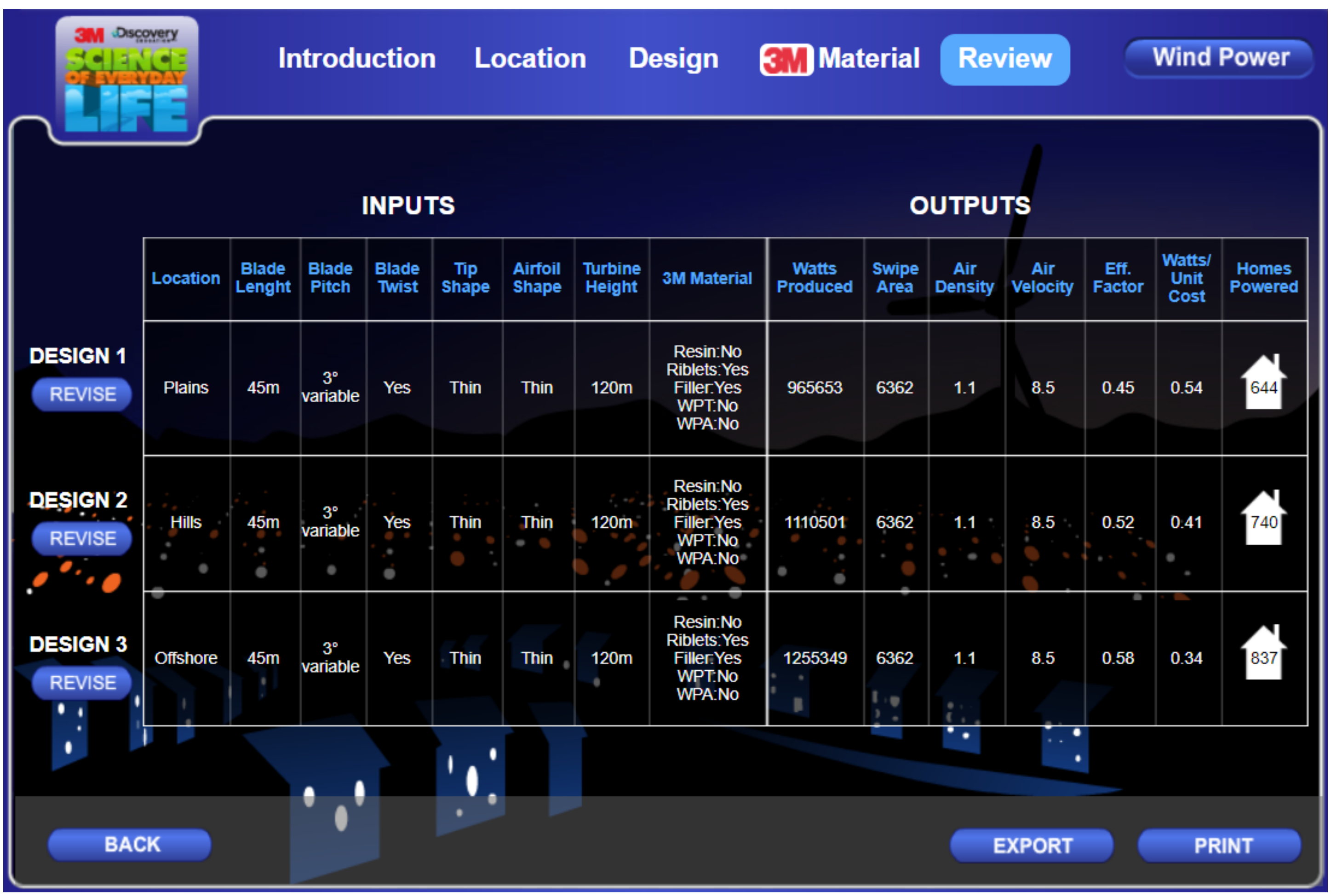
-
Location:Plains
Blade Length:45m
Blade Pitch:3° variable
Blade Twist:Yes
Tip Shape:Thin
Airfoil Shape:Thin
Turbine Height:120m
Resin:No
Riblets:Yes
Filler:Yes
WPT:No
WPA:Yes
Watts Produced:965653
Swipe Area:6362
Air Density:1.1
Air Velocity:8.5
Efficiency Factor:0.45
Watt / Unit Cost:0.55
Homes Powered:644

Design 2

-
Location:Hills
Blade Length:45m
Blade Pitch:3° variable
Blade Twist:Yes
Tip Shape:Thin
Airfoil Shape:Thin
Turbine Height:120m
Resin:No
Riblets:Yes
Filler:Yes
WPT:No
WPA:No
Watts Produced:1110501
Swipe Area:6362
Air Density:1.1
Air Velocity:8.5
Efficiency Factor:0.52
Watt / Unit Cost:0.41
Homes Powered:740

Design 3

-
Location:Offshore
Blade Length:45m
Blade Pitch:3° variable
Blade Twist:Yes
Tip Shape:Thin
Airfoil Shape:Thin
Turbine Height:120m
Resin:No
Riblets:Yes
Filler:Yes
WPT:No
WPA:No
Watts Produced:1255349
Swipe Area:6362
Air Density:1.1
Air Velocity:8.5
Efficiency Factor:0.58
Watt / Unit Cost:0.34
Homes Powered:837



	Location	Blade Lenght	Blade Pitch	Blade Twist	Tip Shape	Airfoil Shape	Turbine Height	3M Material	Watts Produced	Swipe Area	Air Density	Air Velocity	Eff. Factor	Watts/ Unit Cost	Homes Powered
DESIGN 1	Plains	45m	3° variable	Yes	Thin	Thin	120m	Resin:No Riblets:Yes Filler:Yes WPT:No WPA:No	965653	6362	1.1	8.5	0.45	0.54	644
DESIGN 2	Hills	45m	3° variable	Yes	Thin	Thin	120m	Resin:No Riblets:Yes Filler:Yes WPT:No WPA:No	1110501	6362	1.1	8.5	0.52	0.41	740
DESIGN 3	Offshore	45m	3° variable	Yes	Thin	Thin	120m	Resin:No Riblets:Yes Filler:Yes WPT:No WPA:No	1255349	6362	1.1	8.5	0.58	0.34	837

(https://canvas.instructure.com/courses/22 2/40 Graded 17.5 / 20 (88%) Average 40/40 ASWIN VR

Submitted: Oct 18 at 8:35pm
Student Viewed Document: Oct 18 at 8:35pm

Submitted Files: (click to load)

RSE Assignment 2 953617114012.docx

Assessment

Grade out of 20

19

View Rubric

Some Rubric		
Criteria	**Ratings**	
Timely Submission	**Excellent** Submitting on or before time **Comments** Appreciations for your timely submission Aswin	4 / 4 pts
Acknowledgement	**No details** **Comments** The reference link of the site is available in your screenshot, but you have missed out the acknowledgement	2 / 3 pts
Content of the assignment	**Excellent** Impressive, unique content with proper screenshot of attendance or output **Comments** Your content is quite impressive and easy to understood what you have performed.	10 / 10 pts
Presentation	**Excellent** Impressive presentation with properly aligned content along with title page **Comments** Presentation is also quite impressive	3 / 3 pts
		Total Points: 19

Assignment Comments

Add a Comment

Submit

Download Submission Comments